

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_f9aa1d1c-c83\agent-a56ae22f374afc317.jsonl`  
- SHA-256: `75ab2abb5833e45f6c037020c793e179670bf4ffc12174f877abdc47b9ff5a91`  
- Source modified: `2026-06-09T11:30:11+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `wf_f9aa1d1c-c83`  
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-09T11:25:35.354Z)

DIMENSION = FINE-TUNING FRAMEWORKS for video VLMS (SFT + RFT/GRPO for temporal grounding + conversational QA). Use WebSearch+WebFetch. Compare ms-swift, LLaMA-Factory, axolotl, unsloth, HF transformers+trl. CLARIFY for the owner that a framework (ms-swift) and a base model (Gemma/Qwen) are ORTHOGONAL ? you use a framework to fine-tune a base. Which framework best supports native VIDEO + AUDIO training, timestamp/grounding outputs, RFT with a custom reward, and runs on rented A100/L40S? License of each (must be permissive).

OWNER CONTEXT:

OWNER'S REFRAMED VISION (from PR #61 review comments ? supersedes earlier framing where they differ):

- IMMEDIATE GOAL: a TRAINING FRAMEWORK + HARNESS to produce a VERY HIGH precision/recall model that extracts

- HIGHLIGHTS and RALLIES from a clip. Quality first. On-device is LATER (a model-compression goal for power users to

- save upload cost) ? NOT the immediate goal; don't over-index on it now.

- MULTISPORT FROM THE START: all NET-SEPARATED RACQUET SPORTS (badminton, tennis, table tennis, pickleball, padel)

- first, eventually ALL sports. A SINGLE, sports-AGNOSTIC model (the collector + indexer are already sport-generic by

- design; the repo will be renamed off "badminton"). More labels/compute/storage is an accepted "good problem."

- CONVERSATIONAL VIDEO-QA is a north-star feature: upload a video -> the model processes it -> the user asks CONTEXTUAL

- QUESTIONS continuously, e.g. "show me the longest rally", "make a video of all the service faults in this clip",

- "show me a clip where the rally lasted over 20 shots". This implies per-video

STATE/bookkeeping the model itself

- won't hold ? so the FRAMEWORK must be designed for that bookkeeping from the start. Owner asks: feasible to embark on

- now, or punt and revisit after quality is solid?

- HARD COMMERCIAL-LICENSE GUARDRAIL (owner: "never corrupted"): EVERY dependency ? code, data, AND model weights ? must

- be usable in PRIVATE PROPRIETARY COMMERCIAL software. MIT / Apache-2.0 / BSD are the easy fits. Reject viral/NC/

- custom-restrictive (e.g. Gemma-3 viral terms; weights trained on paid-LLM outputs; >X-MAU caps if truly binding).

- NEVER train on paid-LLM (Gemini/OpenAI/Anthropic) outputs. Exclude minors; separate training opt-in. Hopefully a

- CONVERSATIONAL single model.

- CURRENT STATE (de-risking evidence): on a 6-match human-labeled corpus, a learned per-frame prototype gets held-out

- AUC 0.83-0.92 and a cross-video LOVO mean 0.583 vs the heuristic's 0.480 ? and the learned model DECOUPLES from the

- multi-court "background games" confound the heuristic can't escape. WASB = the shuttle

detector (MIT code, but the

distributed checkpoint is academic-data-trained ? provenance unresolved = a commercial-cleanliness blocker to fix).

- CONSTRAINTS: solo founder, Bangalore; local dev box = ONE RTX 2070 Super (8GB) + WSL2; paid Gemini stays the fallback;
cloud GPU rented per-job (Modal/RunPod per the earlier study).

REQUIREMENTS BRIEF:

{"requirements":["GOAL (reframed): The immediate deliverable is a TRAINING FRAMEWORK + HARNESS that produces a VERY high precision/recall model for extracting HIGHLIGHTS and RALLIES from a clip. Quality is the first-order objective; everything else is secondary to model quality.", "ON-DEVICE IS DEFERRED (reframe vs earlier 'on-device = immediate moat'): On-device / model-compression is a LATER goal, motivated as an upload-cost saver for power users. It is NOT the immediate target and the strategy must NOT over-index on it now. (Supersedes OWNED_MODEL_IMPLEMENTATION_PLAN.md's 'on-device = the privacy moat / immediate' framing.)", "MULTISPORT FROM DAY ONE: Build for ALL net-separated racquet sports first (badminton, tennis, table tennis, pickleball, padel), with a path to eventually ALL sports. No single-sport (badminton-only) scoping.", "SINGLE SPORT-AGNOSTIC MODEL: Target ONE sports-agnostic model rather than per-sport models. Collector + indexer are already sport-generic by design; the repo will be RENAMED off 'badminton'. Generic data schema stays; coach->athlete remains an optional overlay, not baked in.", "SCALE COST ACCEPTED: More labels / compute / storage from multisport breadth is an explicitly accepted 'good problem' ? do not constrain the design to minimize label/compute/storage volume at the expense of breadth or quality.", "CONVERSATIONAL VIDEO-QA IS THE NORTH-STAR FEATURE: upload a video -> model processes it -> user asks CONTEXTUAL questions CONTINUOUSLY (e.g. 'show me the longest rally', 'make a video of all the service faults', 'show me a clip where the rally lasted over 20 shots'). Hopefully served by a single conversational model.", "PER-VIDEO STATE/BOOKKEEPING IS A FRAMEWORK REQUIREMENT: The conversational-QA loop implies durable per-video state (rally inventory, shot counts, event/fault tags, timestamps, derived clips) that the MODEL ITSELF will not hold. The FRAMEWORK must be DESIGNED FOR THIS BOOKKEEPING FROM THE START, even if the QA feature ships later.", "OPEN DECISION the owner is asking for a recommendation on: is conversational-QA feasible to EMBARK ON NOW, or should it be PUNTED and revisited AFTER rally/highlight quality is solid? (The bookkeeping-from-the-start requirement holds either way.)", "HARD COMMERCIAL-LICENSE GUARDRAIL ('never corrupted'): EVERY dependency ? code, training data, AND model weights ? must be usable in PRIVATE, PROPRIETARY, COMMERCIAL software. MIT / Apache-2.0 / BSD are the accepted easy fits.", "REJECT viral / non-commercial / custom-restrictive licenses: explicitly reject Gemma-3 viral terms, weights trained on paid-LLM outputs, and >X-MAU caps IF truly binding for commercial use.", "NEVER TRAIN ON PAID-LLM OUTPUTS: Gemini / OpenAI / Anthropic outputs must never be a training-data source (terms/copyright). Paid Gemini remains inference/serving/FALLBACK only. (Also: the existing Gemini-silver training labels in training.label_candidates must be purged before they bake in ? M0.3.)", "DATA-ELIGIBILITY GUARDRAILS: Exclude minors from the training pool; per-user training data requires a SEPARATE explicit opt-in. Split data by match.", "QUALITY EVIDENCE / STARTING POINT (de-risking baseline to beat): On a 6-match human-labeled corpus, the learned per-frame prototype achieves held-out AUC 0.83-0.92 and cross-video LOVO mean 0.583 vs the heuristic's 0.480. The learned model is the chosen path BECAUSE it DECOUPLES from the multi-court 'background games' confound that the heuristic cannot escape (3 heuristic levers ? spatial gate, audio-foreground, court-region separability ? were each tested and failed this session; the problem is temporal, not spatial).", "WASB PROVENANCE IS AN OPEN COMMERCIAL-CLEANLINESS BLOCKER: WASB is the shuttle/ball detector (MIT code), but the distributed checkpoint is academic-data-trained with unresolved provenance (the C3 item) ? this is a commercial-cleanliness blocker that must be resolved (document as Gen-0 prerequisite + plan own-weights, or scope an alternative.)", "RESOURCE/OPERATING CONSTRAINTS: Solo founder, Bangalore. Local dev box = ONE RTX 2070 Super (8GB VRAM) + WSL2 (no full-pipeline run in the dev container ? no GPU/torch/cv2/numpy there). Cloud GPU rented per-job (Modal / RunPod per the earlier study). Paid Gemini stays the fallback.", "LANGUAGE/SCOPE NOTE: The reframed vision adds no new natural-language/multilingual requirement beyond English conversational-QA phrasing; 'language' as a strategic axis is limited to the conversational-QA query interface, not localization.", "tensions":["QUALITY-FIRST vs ON-DEVICE-LATER vs HARDWARE: The very-high-quality goal pushes toward larger VLM bases and cloud training, but the only local box is an 8GB RTX 2070 Super and on-device is explicitly deferred ? so the immediate model is NOT constrained to fit 8GB, yet the eventual compression goal must remain reachable. The strategy must pick a base that is both high-quality-trainable AND later-compressible without re-architecting.", "MULTISPORT-FROM-START vs CURRENT EVIDENCE DEPTH: The de-risking corpus is badminton-only (WASB shuttle features, multi-court badminton confound). A single sport-agnostic model is required from day one, but the only validated signal/features and

the only labeled corpus are badminton ? the feature stack (WASB trajectory) may not generalize to tennis/table-tennis/padel, and per-sport corpora don't exist yet.", "SINGLE SPORT-AGNOSTIC MODEL vs COMMERCIAL-CLEAN DETECTORS: A sport-agnostic model likely needs per-sport object/ball detectors or a unified detector; WASB (badminton) already has unresolved provenance (C3), and clean MIT/Apache/BSD detectors for tennis/TT/pickleball/padel are not yet identified ? each new sport reopens the license-cleanliness guardrail.", "CONVERSATIONAL-QA NORTH-STAR vs 'NEVER TRAIN ON PAID-LLM': A single conversational model that answers contextual video questions typically wants an LLM/VLM teacher, but the strongest available teachers (Gemini/OpenAI/Anthropic) are forbidden as a training source ? so the conversational capability must come from an Apache-2.0/MIT base + own data, narrowing the candidate set and raising the quality bar to clear without paid-LLM distillation.", "BOOKKEEPING-FROM-THE-START vs SEQUENCING DISCIPLINE: The owner wants the framework designed for per-video QA state NOW, but the committed next slice is the async job model (single-tenant) and heavy model work is gated on scaffolding + sign-off ? designing durable QA-state bookkeeping prematurely risks building against an architecture (and a model) that doesn't exist yet (the 'embark now vs punt' question the owner is explicitly asking).", "ACCEPTED SCALE COST vs SOLO-FOUNDER / SINGLE-GPU BUDGET: 'More labels/compute/storage is a good problem' collides with a solo founder on one 8GB GPU renting cloud per-job ? multisport label volume and storage are accepted in principle but still bounded by one person's labeling throughput and per-job cloud spend.", "'SINGLE CONVERSATIONAL MODEL' (hoped) vs HARNESS-FIRST QUALITY GOAL: The owner hopes for one model that does detection + segmentation + conversational QA, but the harness/quality goal is concretely about a rally/highlight extractor; folding conversational QA into the same model vs keeping extraction (model) and QA (bookkeeping + retrieval/LLM layer) separate is an unresolved architectural fork.", "QUALITY-FIRST ('very high precision/recall') vs HONEST EVIDENCE CEILING: Current best cross-video LOVO is ~0.583 (directional, n<=6, linear, threshold-transfer unsolved) ? a large gap from 'very high precision/recall', so the quality target may require substantially more data/features/base-model than the current de-risking artifacts demonstrate."]]

2. assistant (2026-06-09T11:25:38.278Z)

I'll research fine-tuning frameworks for video VLMs. Let me start by loading the web tools and checking any existing project docs on this topic.

3. user (2026-06-09T11:25:39.205Z)

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md

4. assistant (2026-06-09T11:25:41.984Z)

Let me read the existing project study docs first to ground my research in what's already been decided, then run web searches.

5. user (2026-06-09T11:25:42.639Z)

```
1      # Owned-model training study (rally segmentation ? end-to-end multimodal)
2
3      **Date:** 2026-06-08 . **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM
P0?P4 + healthy corpus; owner sign-off required).**
4
5      > How to train our **OWNED** rally/highlight model **reusing the videos we label**,
aligned to the
6      > Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
7      > workflow (ground ? 5 web-research dimensions ? adversarial verification of
costs/licenses/privacy ?
8      > synthesis + completeness critic; 35 sources). Empirical results from THIS session
(LOVO + a learned
9      > prototype) are folded in and **resolve the study's biggest open unknown** ? see §"What
we proved today".
10     > Numbers from web sources are 2026 snapshots in a ±15?25% band; **re-quote live before
budgeting.**
11
12     ## Thesis
```

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13
14     Train the owned model in **three staged generations off ONE model-agnostic labeled
corpus** ? a label
15     collected once feeds every generation, so labels are never re-collected. **The moat is
the consented
16     dataset + the eval harness, never the weights.**
17
18     | Gen | When | Approach | Base / model | Compute |
19     |---|---|---|---|---|
20     | **Gen-0** | now ? unblocks at n?5 matches | tiny **temporal action segmentation
(TAS)** head over a hand-built per-frame feature stack from the WASB trajectory we already
produce; per-frame rally(1)/non-rally(0) ? merge into [start,end] | **ASFormer (MIT, ~1.1M
params)** primary; **ActionFormer (MIT)** A/B challenger (overtakes once corpus = hundreds of
matches) | **local RTX 2070 Super**, trains in minutes, $0 |
21     | **Gen-1 / Option A** | 6?12 mo | concatenate more local cue channels onto the SAME
head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues
swappable | same ASFormer/ActionFormer, wider input | local + occasional cloud burst (pseudo-
labeling) |
22     | **Gen-2 / Option B** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end
multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start
then **GRPO/RFT with a temporal-IoU-vs-golden reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0,
on-device) primary; **Qwen2.5-VL-7B** fallback; **Gemma 4 E4B/12B-Unified** (Apache-2.0 + native
video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |
23
24     ## What we proved today (de-risks the blueprint)
25
26     The study's **"dominant empirical unknown"** was: *can a low-dim WASB trajectory (?3?4
raw dims) feed a
27     TAS head, given ASFormer's small-data result used 2048-d I3D features?* This session
answered it.
28
29     - **Cross-video LOVO of the current heuristic** (`calibrate_local`, Adarsh-96 +
Kushagra-32, human labels):
30     mean held-out F1 **0.443 ± 0.279**; Adarsh 0.722, **Kushagra 0.163** (its own ceiling
? *no* velocity-windowing
31     config does better). **The ±0.279 spread is method variance, not config.** ? This also
**corrects the study's
32     "~0.73 bar": that was a fit-on-self single-match number; the honest cross-video
heuristic bar is **~0.44**,
33     which *lowers* the Gen-0 ship gate.
34     - **A learned per-frame prototype** (`backend/eval/rally_seq_proto.py`: 6 hand-built
features ? visibility rate,
35     mean/max shuttle speed, **net-crossing rate**, x/y spread ? into the repo's
dependency-free LogisticRegression,
36     honest LOVO): **per-frame AUC 0.878 (Adarsh) / 0.841 (Kushagra), held-out.** On
Kushagra ? the match the
37     heuristic *cannot* segment (0.163) ? a calibrated threshold reaches **segment F1 0.578
(3.5x)**. The naive
38     cross-video threshold gives only 0.049, so **the remaining gap is threshold/operating-
point calibration at n=2,
39     not representation.**
40
41     - **n=3 update (doubles match added ? `LargeTestVideo`, 49 rallies):** the learned
prototype **overtakes the
42     heuristic.** Mean held-out LOVO F1 ? learned **0.377 ? 0.615**, heuristic **0.443 ?
0.544**. Kushagra's learned
43     held-out F1 jumped **0.049 ? 0.417** (one added match closed most of the calibration
gap); the **doubles** match,
44     held out and trained only on the two singles, scored AUC **0.914** / F1 0.729
(beneralizes across play type +
45     frame rate). The heuristic still cannot rescue Kushagra at any n (0.151, capped).
**The honest "bar to beat" is
46     thus ~0.54 (n=3), not 0.73 ? and the learned path already clears it on the mean and
keeps rising with n.**
47

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48     **Conclusion:** the rally signal IS learnable from the WASB trajectory features, cross-
footage (AUC ~0.85?0.91) ?
49     the heuristic just can't exploit it. This **validates the Gen-0 ASFormer-over-WASB-
features path**, shows the
50     hand-built feature stack the study calls for is the right lever, and confirms the
bottleneck is **data + per-video
51     calibration (n?5 matches), not method.** The prototype is the seed of Gen-0, and the
n=2?n=3 trend (learned mean
52     +0.24, overtaking the heuristic) is the empirical green light.
53
54     ## Platform ranking (privacy + cost lens; the fine-tune is a sub-24h, $10?50 job ?
privacy/ergonomics dominate)
55
56     1. **Modal ? PRIMARY.** Strongest *written* no-access contract (verbatim: "Modal will
never access? the inputs/outputs to your Functions, or any data you store"), ?7-day TTL,
SOC2-II, HIPAA-BAA; per-second scale-to-zero matches retrain-on-each-batch. A100 80 GB
~$2.50/hr, H100 ~$3.95/hr (corrected down from cited). Privacy is the load-bearing reason, not
price.
57     2. **RunPod (Secure Cloud tier ONLY) ? FALLBACK / Option-B scale-up.** Cheapest vetted-
DC self-serve; A100 80 GB ~$1.39?1.49/hr. Community/P2P tier is UNVETTED ? **never** touches
consented video. Confirm AES-256/DPA at booking.
58     3. **Vertex AI / GCP ? ON BENCH.** Only strong-privacy option with an India region, but
it's **capacity-gated** (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100)
and **tuning Gemini yields NO portable on-device weights** ? only tuning an Apache-2.0 base in
Model Garden does. Use only if India residency-at-rest is mandated.
59     4. **Self-hosted RTX 2070 Super 8 GB** ? Gen-0 training + ALL inference + CI only.
**Cannot** QLoRA a 7B video VLM (~20?24 GB). Best privacy (nothing leaves the box).
60     5. **Fireworks** (managed, contractual no-train + ZDR) usable; **Together BLOCKED**
until a signed DPA + explicit no-train clause. AWS SageMaker / Azure ML = heavy for a solo
founder; India-residency only.
61
62     ## Label reuse ? one corpus, three transcoders (the actual moat)
63
64     Promote the rally CSVs from eval-only files to **one model-agnostic corpus**: JSONL, one
row per
65     `(video_id, rally_ordinal)` = `{start, end, ending_reason, sport, source
(human|pseudo|vendor),
66     annotator/provenance, consent/opt-in, label_type, split (BY MATCH not clip),
trajectory_ref}`. The
67     provenance/consent fields are the **enforcement point** for the legal guardrails. Three
deterministic
68     transcoders read it, zero re-labeling:
69     1. **Heuristic/LogReg view** ? already exists
(`backend/eval/training.py::label_candidates` via `metrics.match_intervals`).
70     2. **TAL view** ? ActivityNet JSON (`segment:[start,end]`, `label:'rally'`); our seconds
map 1:1.
71     3. **VLM clip?QA view** ? "list every rally [start,end] in this clip"; `ending_reason` ?
a second QA task.
72
73     **One scorer spine:** reuse `metrics.match_intervals` for heuristic-labeling, TAL eval,
AND the Gen-2 RFT
74     reward ? no metric drift. **Leakage guard:** split **by match, never by clip**.
**Amplification (all
75     guardrail-clean):** open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never
paid-LLM),
76     boundary-preserving temporal augmentation, active learning to spend the Rooru budget on
highest-uncertainty
77     matches. ? **The new partial-label windowing**
(`eval_partial_labels.restrict_to_window`) must flow into the
78     TAL/VLM views so prefix-labeled videos aren't mis-scored (a corpus that's partially
tagged is the norm).
79
80     ## Licensing & privacy guardrails (all respected; verify per-checkpoint)
81
82     - **Bases: Apache-2.0/MIT only.** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0),

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Qwen3-VL-4B/8B (Apache-2.0),
83 Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B
(MIT/Apache), ASFormer/
84 ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP),
Qwen2.5-VL-3B (NC research),
85 Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3
weights (NC + paid-LLM-trained),
86 TimeChat/VTimeLLM/Momentor (NC/no-license = method-reference only), VideoMAE weights
(CC-BY-NC), `yabufarha/ms-tcn`
87 (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names =
E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
88 - **No paid-LLM training data** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-
reference only, **never** a
89 training signal. Gemini silver labels stay out of every transcoder.
90 - **Exclude minors by default + separate training opt-in;** per-user adapters trained
only on that user's
91 consented video (deletion = drop the adapter). Enforced at the corpus-membership layer
before transcoding.
92 - **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any
cloud *training* needs a
93 signed DPA + no-train assurance; route only owned/consented footage, never user
uploads.
94
95 ## Cost (spend on labels, not GPUs)
96
97 - **Gen-0:** ~\$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).
98 - **Gen-2 LoRA:** ~\$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token
cost on real 1080p/60fps
99 clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together)
~\$2?30 *if* text-priced
100 (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).
101 - **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini
(~\$0.05/10-min clip) **only**
102 above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now,
owned-primary later.
103 **Benchmark \$/call before committing** (COMMERCIALIZATION C7).
104
105 ## Open risks & critic caveats (read before acting)
106
107 - **?? WASB checkpoint provenance (C3) propagates.** The study calls WASB "MIT," but
that's the *code*; the
108 distributed `wasb\_badminton\_best.pth.tar` is **academic-data-trained, provenance
unresolved** (COMMERCIALIZATION
109 C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label
teacher ? a "clean" owned
110 model trained on provenance-unclear features **does not launder the provenance.**
ROADMAP Phase-4 "train own WASB
111 weights" is a **prerequisite, not an aside.**
112 - **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER
until a raw HF LICENSE
113 revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is
less documented than Qwen's.
114 - **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-
only policy now **forbids
115 this; that pipeline must be purged of Gemini-derived training labels** to comply.
116 - **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the
approved list includes MIT ?
117 ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at
our scale.
118 - **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on
8 GB Turing at
119 1080p/60fps. **Measure before relying on it.**
120 - **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k
clip?QA band alone; if
121 augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed.

Amateur 60fps may need more
 122 labels than broadcast-trained literature implies.
 123 - ****Dropped option worth keeping:**** InternVideo2 encoder (MIT) + light grounding head ?
 the ***only*** data-realistic
 124 owned-video experiment available at n?128 today. And: a ****one-time ~\$1?2k 24 GB local**
card** (e.g. used 3090)
 125 would preserve the privacy lever for ***training*** and dissolve most of the cloud-DPA
 analysis ? uncoded here.
 126
 127 **## Open questions (need the GPU/WSL machine-side suite ? not runnable in this dev**
 container)
 128
 129 1. Does ASFormer/ActionFormer over the WASB feature stack beat the ****honest ~0.44 LOVO****
 bar at n?5 matches?
 130 (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now ***likely***;
 confirm at segment level.)
 131 2. Real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on our actual
 clip?QA frame counts?
 132 3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?
 133 4. Measured self-hosted-7B \$/call + latency vs Gemini on the target workload (the owned-
 model ROI crossover)?
 134 5. Will Together/Lambda/Azure sign a DPA with explicit no-train for consented video,
 India-configurable?
 135 6. Realistic clip?QA yield from 128 rallies via augmentation + QA-multiplexing + pseudo-
 labels?
 136
 137 **## Sequencing (owner directive, non-negotiable)**
 138
 139 This is a ****design blueprint only.**** It unblocks PLATFORM ****P5**** but starts ****no**
 implementation** until P0?P4
 140 scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus
 is healthy. Paid Gemini
 141 stays the always-ready primary/fallback. The only items safe to act on now ? ****and even**
 these need owner
 142 sign-off before code** ? are (a) the model-agnostic corpus format and (b) growing n?2 ?
 n?5 matches (a small
 143 Roora batch), which unblocks honest tuning immediately and is the ****highest-ROI spend**
 (labels, not GPUs).**
 144
 145 ***Full agent transcript + 35 sources: workflow `wf\_49aab888-e10` (this session).***
 146

6. assistant (2026-06-09T11:25:46.024Z)

This study mentions GRPO/RFT and Qwen/Gemma bases but doesn't deeply compare the training frameworks themselves. Let me check the implementation plan briefly and then do focused web research on the frameworks.

7. user (2026-06-09T11:25:47.765Z)

docs\archives\checkpoints\2026-06-strategy-foundation\README.md:17:-
****PLATFORM_ARCHITECTURE.md**** ? multi-user "assistant coach" platform: pluggable
 `StorageBackend`/`ComputeBackend` (BYO), generic schema (coach overlay optional), owned video-QA
 model layer (\$5A), training-backend survey + nanochat-style harness, committed legal guardrail,
 per-user-adapter direction, v1-vs-vision scoping.
 docs\archives\checkpoints\2026-06-strategy-foundation\README.md:27:- ****Generic data schema****
 (User ? VideoCollection ? Video ? Highlight ? HighlightMetadata); coach ? athlete is an
****optional overlay**** (a market-fit hypothesis).
 docs\CODE_MAP.md:96:[Omitted long matching line]
 docs\CODE_MAP.md:112:[Omitted long matching line]
 docs\CODE_MAP.md:140:[Omitted long matching line]
 docs\CODE_MAP.md:141:[Omitted long matching line]
 docs\CODE_MAP.md:253:- ****`calibrate\_wasb.{load\_golden\_gts, make\_predict\_fn, BASELINE}`** is a de-
 facto shared API** for calibrate_local, export_wasb_segments, gemini_refine, e1_probe ? changing
 its golden-CSV schema or **`BASELINE`** ripples widely. **`gemini\_refine.merge\_overlaps`** reused by

`distill\_local`; `harness.get\_predictions` reused by `regression.regress`.

docs\CODE_MAP.md:315:| `test\_gemini\_refine.py` | `eval.gemini\_refine` merge_overlaps / _offset_segments (pure helpers, no API) |

docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:211:that overlap the foreground image area. ? **Court-region detection (heuristic OR a dedicated model) would NOT

docs\DEFERRED_HARDENING_PLAN.md:74: - `assert\_disjoint(train\_ids, eval\_ids)` ? raise on any overlap.

docs\DEFERRED_HARDENING_PLAN.md:85:- Unit test: overlapping train/eval ids ? `assert\_disjoint` raises.

docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:198:- **Items 4 (logging), 5 (packaging), 6 (detector runner abstraction) ? safe to do independently** (near-zero overlap with the reports). For #5, model `pyproject.toml` on `obsreport`'s and pin the `obsreport` dep tag.

docs\EVALUATION.md:91: overlap their ground truth).

docs\EVALUATION.md:114:**matches** a ground-truth rally when their **temporal IoU** (overlap ÷ union) is

docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:58:| **E1** | Extract audio ? energy-peak hits ? overlay hit-density on WASB windows **and** golden labels | audio + ?1 golden-rated video | **GO/NO-GO** : do hit-clusters align with real rallies + recover WASB-missed ones on *real GoPro* audio? |

docs\archives\decisions\DECISIONS.md:180: `yolo\_learned`): label detector windows by overlap with GT rallies, train a lightweight

docs\archives\decisions\DECISIONS.md:205:- **72%** of rallies have *some* candidate overlap, yet the **median best coverage of a

docs\archives\research\RALLY_DETECTION_RESEARCH.md:44:- Scoreboard/score-overlay OCR: usable only when a stable on-screen score exists ? mostly N/A for amateur clips.

docs\archives\research\RALLY_DETECTION_RESEARCH.md:58: impact timestamps ? overlay hit density on existing WASB windows ? does audio recover rallies WASB misses?

docs\HOW_RALLY_DETECTION_WORKS.md:46:rallies with **temporal IoU** (how much they overlap), and pick the 3 knobs that maximise it ?

docs\PLATFORM_ARCHITECTURE.md:30:athlete overlay over a generic core** (§4.1) ? not baked in; (2) gait/biometrics is

docs\PLATFORM_ARCHITECTURE.md:176:### 4.1 Entities ? generic core, with the coach model as an optional overlay

docs\PLATFORM_ARCHITECTURE.md:193:- **Coach overlay (optional, additive):** for the academy/coach GTM, layer

docs\PLATFORM_ARCHITECTURE.md:199: overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g. shot

docs\PLATFORM_ARCHITECTURE.md:336:as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*

docs\PLATFORM_ARCHITECTURE.md:343:| **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio** ; Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | **? video-QA; slots into BYO-compute (§3.2)** |

docs\PLATFORM_ARCHITECTURE.md:344:| **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B | ? | ? | popular, well-documented |

docs\PLATFORM_ARCHITECTURE.md:345:| **Unsloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) | any incl 7B | ? GRPO | ? | cheapest first runs |

docs\PLATFORM_ARCHITECTURE.md:356:- **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /

docs\PLATFORM_ARCHITECTURE.md:357: Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud): native

docs\PLATFORM_ARCHITECTURE.md:388:**Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,

docs\PLATFORM_ARCHITECTURE.md:390:`args.json` + `full\_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?

docs\PLATFORM_ARCHITECTURE.md:397:2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an orchestration

docs\PLATFORM_ARCHITECTURE.md:404:So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register

docs\PLATFORM_ARCHITECTURE.md:406:wrapper; ms-swift is the engine; the **golden QA eval is the report card**.

docs\PLATFORM_ARCHITECTURE.md:493: where the overlay applies, parent?minor).

docs\PLATFORM_ARCHITECTURE.md:561:[Omitted long matching line]

docs\PLATFORM_ARCHITECTURE.md:563:[Omitted long matching line]

8. assistant (2026-06-09T11:25:51.638Z)

There's prior work in PLATFORM_ARCHITECTURE.md that already leans toward ms-swift. Let me read that section to understand the existing position, then verify/extend with current web research.

9. user (2026-06-09T11:25:52.542Z)

```
325      cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
326      is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
327
328      Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
329
330      ### Training backend ? managed service vs self-managed (the fine-tune "forge")
331      *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
export/ownership,
332      smallest servable model, RL support, no-train DPA, and infra burden**.
333
334      **Decisive finding (researched 2026-06):** managed fine-tuning services are
335      **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
336      as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
337      video (+audio) training.** So for true video-QA, a self-managed framework is the
338      strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
339      serve) and keep full ownership. Managed services trade some of that for zero infra.
340
341      | Option | Type | Video / VLM support | Smallest VL | RL | Export weights | Best for |
342      |---|---|---|---|---|---|---|
343      | **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio**;
Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | **? video-QA;
slots into BYO-compute ($3.2)** |
344      | **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B
| ? | ? | popular, well-documented |
345      | **Unsloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) |
any incl 7B | ? GRPO | ? | cheapest first runs |
346      | **Tinker** (Thinking Machines) | Managed API (low-level) | Qwen3-VL **30B-A3B+ only**;
image input (video **unverified**) | 30B-A3B MoE | ? PPO/GRPO, custom loss | ? adapters + ckpt;
no-train ToS | zero-infra RL/research at MoE scale |
347      | **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data |
verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |
348      | **Fireworks AI** | Managed (tune + serve) | Full Qwen2.5-VL (3B?72B SFT; LoRA@32B +
LoRA-upload) | **7B** | partial | ?? verify export (lock-in risk) | managed *small*-VL + serving
|
349      | **Predibase** (Rubrik) | Managed **+ VPC self-host** | VLM instruct-tune; **LoRAX**
multi-adapter | verify | ? (VPC) | ? **RFT/GRPO** flagship | RL "creativity" + multi-adapter +
privacy |
350      | ~~OpenAI GPT-4o / Gemini tuning~~ | Closed | ? | ? | ? | ? no export | **non-fit ?
violates "own the model"*** |
351
352      *(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
353      control but are only worth it if you're already on that stack.)*
354
355      **Recommendation:**
356      - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
357      Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):
native
358      video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
359      **ComputeBackend / BYO-compute** seam ($3.2). **Likely a better fit than Tinker** for
360      the video requirement *and* serving economics.
361      - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
362      or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
363      adapter export.
364      - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
365      (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
366      level RL primitives).
```

```

367 - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just
368 not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
369 - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
370 backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.
371
372 **Verify before committing (any backend):** (1) **native video / frame-sequence**
373 ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering
374 consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the
size
375 you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).
376
377 ### Training harness ? make it agent-runable (nanochat-style operability)
378 The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video
379 and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is
the
380 reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
381 tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
382 score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
383 five properties are exactly what make training agent-runable, and we copy them:
384 **(1)** one command, end-to-end; **(2)** one complexity dial (sane auto-derived
defaults);
385 **(3)** an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
386 **(4)** reproducible + deterministic + cheap; **(5)** minimal/readable.
387
388 **Engine (don't reinvent):** **`ms-swift`** already delivers this for VLMs ? CLI-driven,
389 **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
390 `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
391 ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)
392
393 **The harness = thin glue over components we already have** ? one declarative,
394 agent-invokable job (call it `train_speedrun`):
395 1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
396 `candidate_segments` + golden set ? the labeling seam in `backend/`).
397 2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an
orchestration
398 job (§5 / DEFERRED #16): one config, idempotent, logged.
399 3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to

```

10. assistant (2026-06-09T11:25:57.093Z)

I have a strong grounding now. The repo already favors ms-swift but the comparison is from an earlier framing and predates the reframed vision (RFT with custom reward, conversational QA, audio). Let me run focused current web research to validate and update each framework's status, especially on: native video+audio, GRPO/RFT with custom rewards, timestamp grounding, and licenses.

11. user (2026-06-09T11:26:06.374Z)

Web search results for query: "LLaMA-Factory video VLM fine-tuning GRPO reward function 2025 2026"

Links: [{"title": "LLaMA-Factory/README.md at main · hiyouga/LLaMA-Factory", "url": "https://github.com/hiyouga/LLaMA-Factory/blob/main/README.md"}, {"title": "Fine-Tuning LLaMA for Beginners: A 2026 Step-by-Step Guide", "url": "https://www.zignuts.com/blog/fine-tuning-llama-for-beginners"}, {"title": "What is LLaMA Factory? LLM Fine-Tuning | VoltAgent", "url": "https://voltagent.dev/blog/llama-factory/"}, {"title": "ViLaCD-R1: A Vision-Language Framework for Semantic Change Detection in Remote Sensing", "url": "https://arxiv.org/pdf/2512.23244"}, {"title": "LLaMA Factory: Complete Guide to LLM Fine-Tuning in 2026", "url": "https://www.jenova.ai/en/resources/llama-factory-complete-guide-"}]

```
to-llm-fine-tuning"},"title":"GitHub - hiyouga/LlamaFactory: Unified Efficient Fine-Tuning of 100+ LLMs & VLMs (ACL 2024) · GitHub","url":"https://github.com/hiyouga/LlamaFactory"},"title":"Chain-of-Zoom: Extreme Super-Resolution via Scale Autoregression and Preference Alignment","url":"https://arxiv.org/pdf/2505.18600"},"title":"See2Refine: Vision-Language Feedback Improves LLM-Based eHMI Action Designers","url":"https://arxiv.org/pdf/2602.02063"},"title":"Boosting RL-Based Visual Reasoning with Selective Adversarial Entropy Intervention","url":"https://arxiv.org/pdf/2512.10414"},"title":"Welcome to LLaMA Factory!","url":"https://llamafactory.readthedocs.io/en/latest/"]]
```

Based on the search results, here's what I found about LLaMA-Factory, video VLM fine-tuning, and GRPO reward functions in 2025-2026:

LLaMA-Factory and GRPO Support

LLaMA-Factory announced EasyR1 in February 2025, an efficient, scalable and multi-modality RL training framework for efficient GRPO training. LLaMA Factory is an open-source framework for fine-tuning large language models (LLMs) and vision-language models (VLMs) on custom datasets, supporting supervised fine-tuning, RLHF, DPO, and other training methods across 100+ model architectures.

VLM Fine-tuning Capabilities

LLaMA-Factory supports fine-tuning vision-language models including the DeepSeek-R1 and Qwen2.5-VL models, as well as MiniCPM-o-2.6 and MiniCPM-V-2.6 models. Earlier 2025 releases added support for GLM-4.1V, Qwen3, InternVL3, Llama 4, and Qwen2.5-Omni, with support including audio models and multimodal models.

GRPO Reward Function

GRPO is a value-model-free RL algorithm widely used to improve the reasoning capacity of VLMs through finetuning, getting rid of the value model in PPO by utilizing rule-based reward function. The reward function mainly consists of accuracy reward and format reward?the former evaluates if the response is accurate while the latter enforces the model to output structured responses.

2026 Outlook

By 2026, GRPO will often be used, which allows the model to "reason" through a problem multiple times, rewarding the paths that lead to the most accurate and logical conclusion.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. user (2026-06-09T11:26:06.890Z)

Web search results for query: "ms-swift ModelScope video training Qwen3-VL GRPO custom reward 2026"

Links: [{"title":"MS-SWIFT - Qwen","url":"https://qwen.readthedocs.io/en/latest/training/ms_swift.html"},"title":"GitHub - modelscope/ms-swift: Use PEFT or Full-parameter to CPT/SFT/DPO/GRPO 600+ LLMs (Qwen3.6, DeepSeek-V4, GLM-5.1, InternLM3, Llama4, ...) and 300+ MLLMs (Qwen3-VL, Qwen3-Omni, InternVL3.5, Ovis2.5, GLM4.5v, Gemma4, Llava, Phi4, ...) (AAAI 2025). · GitHub","url":"https://github.com/modelscope/ms-swift"},"title":"Qwen3-VL-32B-Instruct GRPO Training Script · Issue #7324 · modelscope/ms-swift","url":"https://github.com/modelscope/ms-swift/issues/7324"},"title":"? Best Practices for Training Qwen3/Qwen3-MoE · Issue #4030 · modelscope/ms-swift","url":"https://github.com/modelscope/ms-swift/issues/4030"},"title":"Question: Multi-frame VL training OOM with Megatron + GRPO (Qwen3-VL-8B) · Issue #7326 · modelscope/ms-swift","url":"https://github.com/modelscope/ms-swift/issues/7326"},"title":"Qwen3 Best Practices ? swift 4.3.0.dev0

```
documentation", "url": "https://swift.readthedocs.io/en/latest/BestPractices/Qwen3-Best-Practice.html"}, {"title": "ms-swift · PyPI", "url": "https://pypi.org/project/ms-swift/"}, {"title": "Qwen/Qwen3-235B-A22B · [Fine-tuning] Qwen3-MoE Megatron Training Implementation and Best Practices?", "url": "https://huggingface.co/Qwen/Qwen3-235B-A22B/discussions/6"}, {"title": "https://github.com/modelscope/ms-swift | Ecosyste.ms: Awesome", "url": "https://awesome.ecosyste.ms/projects/github.com/modelscope/ms-swift"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.9.3 documentation", "url": "https://swift.readthedocs.io/en/v3.9/BestPractices/GRPO-Multi-Modal-Training.html"}]
```

Based on the search results, here's what I found about ms-swift, ModelScope video training, Qwen3-VL, and GRPO with custom rewards in 2026:

Overview

MS-SWIFT is a large model and multimodal large model fine-tuning and deployment framework provided by the ModelScope community. It now supports training (pre-training, fine-tuning, human alignment), inference, evaluation, quantization, and deployment for 600+ text-only large models and 400+ multimodal large models.

Qwen3-VL Support

Multimodal large models include: Qwen3-VL, Qwen3-Omni, Llava, InternVL3.5, MiniCPM-V-4, Ovis2.5, GLM4.5-V, DeepSeek-VL2, etc.

GRPO Training and Custom Rewards

MS-swift integrates numerous GRPO algorithm family reinforcement learning algorithms including: GRPO, DAP0, GSP0, SAP0, CISPO, RL00, Reinforce++, etc.

For custom reward functions, the framework supports custom reward functions through external plugins, as demonstrated in training examples using parameters like `--external_plugins` and `--reward_funcs`

Video Training for Qwen3-VL

There is evidence of video training support. Multi-frame visual input training (max_frames=180) using Megatron + GRPO is being experimented with the Qwen3-VL-8B model.

Recent Updates

MS-swift v4.0 major version was officially released on 2026.03.03.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. assistant (2026-06-09T11:26:12.426Z)

Let me research the remaining frameworks (axolotl, unsloth, TRL) and the specific capabilities that matter: native audio training, timestamp/temporal grounding, and the RFT-with-custom-reward path for video.

14. user (2026-06-09T11:26:20.483Z)

Web search results for query: "HuggingFace TRL GRPO vision language model VLM reward function video 2026"

Links: [{"title": "Post training a VLM for reasoning with GRPO using TRL · Hugging Face", "url": "https://huggingface.co/learn/cookbook/fine_tuning_vlm_grpo_trl"}, {"title": "trl/docs/source/grpo_trainer.md at main · huggingface/trl", "url": "https://github.com/huggingface/trl/blob/main/docs/source/grpo_trainer.md"}, {"title": "GRPO Trainer · Hugging"}]

```
Face", "url": "https://huggingface.co/docs/trl/v0.21.0/grpo_trainer"}, {"title": "GitHub -  
huggingface/trl: Train transformer language models with reinforcement learning. ·  
GitHub", "url": "https://github.com/huggingface/trl"}, {"title": "VLM-R1: A Stable and Generalizable  
R1-style Large Vision-Language  
Model", "url": "https://arxiv.org/pdf/2504.07615"}, {"title": "Releases ·  
huggingface/trl", "url": "https://github.com/huggingface/trl/releases"}, {"title": "Fine-Tuning a  
Vision Language Model with TRL using MPO · Hugging  
Face", "url": "https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_mpo"}]
```

Based on the search results, here's what I found about HuggingFace TRL GRPO for vision language models:

GRPO for Vision Language Models (VLMs)

HuggingFace's Transformer Reinforcement Learning library (TRL) supports post-training Vision Language Models (VLMs) using GRPO for adding reasoning capabilities.

Key Features:

****Reward Functions:**** GRPOTrainer supports multiple reward functions of different types, and the reward will be computed as the sum of the rewards from each function, or the weighted sum if reward_weights is provided in the config.

****VLM Training Support:**** GRPO supports training Vision-Language Models (VLMs) on multimodal datasets containing both text and images. The trainer automatically handles image-to-tensor conversion via the model's image processor.

****Implementation Example:**** A practical example uses Qwen2.5-VL-3B-Instruct with a dataset that includes images with problem descriptions along with their solution and thinking trace, leveraging GRPO reward functions to teach the model how to reason.

****Algorithm Details:**** GRPOTrainer implements the Group Relative Policy Optimization (GRPO) algorithm that is more memory-efficient than PPO and was used to train Deepseek AI's R1.

Note: The search results focus on image-based VLM training. While your query mentioned "video," the search results primarily cover image and text multimodal training with GRPO.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. user (2026-06-09T11:26:20.485Z)

Web search results for query: "unsloth video VLM Qwen3-VL GRPO vision reinforcement learning 2026"

Links: [{"title": "Qwen3-VL: How to Run Guide | Unsloth Documentation", "url": "https://unsloth.ai/docs/models/qwen3-how-to-run-and-fine-tune/qwen3-vl-how-to-run-and-fine-tune"}, {"title": "Vision Reinforcement Learning (VLM RL) | Unsloth Documentation", "url": "https://unsloth.ai/docs/get-started/reinforcement-learning-rl-guide/vision-reinforcement-learning-vlm-rl"}, {"title": "Vision Reinforcement Learning", "url": "https://www.unsloth.ai/blog/vision-rl"}, {"title": "Unsloth: Train Massive LLMs on Consumer GPUs with 70% Less VRAM - BrightCoding", "url": "https://blog.brightcoding.dev/2026/02/05/unsloth-train-massive-llms-on-consumer-gpus-with-70-less-vram"}, {"title": "Unsloth Announces Update: Free Reinforcement Learning Training for Vision Large Model Qwen2.5-VL-7B | by AI Engineering | Medium", "url": "https://ai-engineering-trend.medium.com/unsloth-announces-update-free-reinforcement-learning-training-for-vision-large-model-qwen2-5-vl-7b-9ca961f94743"}, {"title": "Qwen3.5 Fine-tuning Guide | Unsloth Documentation", "url": "https://unsloth.ai/docs/models/qwen3.5/fine-tune"}, {"title": "Qwen3 - How to Run & Fine-tune | Unsloth Documentation", "url": "https://unsloth.ai/docs/models/qwen3-how-to-run-and-fine-tune"}, {"title": "Unsloth - Qwen - Read the Docs", "url": "https://qwen.readthedocs.io/en/latest/training/unsloth.html"}]

Based on the search results, here's what I found about Unsloth's vision reinforcement learning capabilities with Qwen3-VL and GRPO:

Overview

Unsloth now supports vision/multimodal RL with Qwen3-VL, Gemma 3 and more.

Performance Improvements

Unsloth makes VLM RL 1.5-2x faster, uses 90% less VRAM, and enables 15x longer context lengths than FA2 setups, with no accuracy loss.

GRPO and GSPO Support

Unsloth supports Qwen3-VL fine-tuning and RL. This update also introduces Qwen's GSPO algorithm. Enabling GSPO is simple, all you need to do is set the `importance_sampling_level = "sequence"` flag in the GRPO config.

Practical Training

Unsloth can train Qwen3-VL-8B with GSPO/GRPO on a free Colab T4 GPU. The goal of the GRPO notebook is to make a vision language model solve maths problems via RL given an image input.

Model Capabilities

Qwen3-VL has vision, video and OCR capabilities as well as 256K context (can be extended to 1M).

The search results show that Unsloth has made significant advances in making vision language model reinforcement learning more accessible and efficient for researchers and developers.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. user (2026-06-09T11:26:20.544Z)

Web search results for query: "axolotl multimodal vision language model video fine-tuning support 2026"

Links: [{"title": "Axolotl", "url": "https://docs.axolotl.ai/"}, {"title": "Axolotl vs Unsloth vs TorchTune: Best LLM Fine-Tuning Frameworks in 2026 | Spheron Blog", "url": "https://www.spheron.network/blog/axolotl-vs-unsloth-vs-torchtune/"}, {"title": "Axolotl AI", "url": "https://docs.axolotl.ai/index.html"}, {"title": "EVAL #003: Fine-Tuning in 2026 - Axolotl vs Unsloth vs TRL vs LLaMA-Factory - DEV Community", "url": "https://dev.to/ultraduneai/eval-003-fine-tuning-in-2026-axolotl-vs-unsloth-vs-trl-vs-llama-factory-2ohg"}, {"title": "axolotl · PyPI", "url": "https://pypi.org/project/axolotl/"}, {"title": "GitHub - leeroopedia/workflow-axolotl-ai-cloud-axolotl-multimodal-vision-finetuning", "url": "https://github.com/leeroopedia/workflow-axolotl-ai-cloud-axolotl-multimodal-vision-finetuning"}, {"title": "Axolotl: The Production LLM Fine-Tuning Framework You Didn't Know You Needed | Starlog", "url": "https://starlog.is/articles/llm-engineering/axolotl-ai-cloud-axolotl/"}, {"title": "Comparing Fine Tuning Frameworks", "url": "https://www.hyperbolic.ai/blog/comparing-finetuning-frameworks"}, {"title": "GitHub - axolotl-ai-cloud/axolotl: Go ahead and axolotl questions · GitHub", "url": "https://github.com/axolotl-ai-cloud/axolotl"}, {"title": "MultiModal / Vision Language Models (BETA) ? Axolotl", "url": "https://docs.axolotl.ai/docs/multimodal.html"}]

Based on the search results, here's what I found about Axolotl's multimodal vision language model video fine-tuning support in 2026:

Video Fine-Tuning Support

Axolotl supports fine-tuning vision-language models (VLMs) including LLaMA-Vision, Qwen2-VL, Pixtral, LLaVA, SmolVLM2, GLM-4.6V, InternVL 3.5, Gemma 3n, and audio models like Voxtral with image, video, and audio support.

Multimodal Support Status in 2026

Fine-tuning vision-language models is no longer a research afterthought, and Axolotl's multimodal support is stable and documented. LLaMA-Vision, Qwen2-VL, and Pixtral can all be fine-tuned on custom image-text datasets.

The framework has evolved significantly?multimodal model fine-tuning support was initially introduced as beta in March 2025, and by 2026 it has become a stable feature. Axolotl now handles LLaMA-Vision, Qwen2-VL, and Pixtral natively.

Training Methods

Axolotl supports various training approaches including LoRA and QLoRA for parameter-efficient fine-tuning, which allows users to fine-tune these multimodal models efficiently even with limited computational resources.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. assistant (2026-06-09T11:26:26.012Z)

I have good coverage. A key signal: TRL's GRPO VLM support is image-only; Unsloth's RL is image-only; only ms-swift explicitly advertises native video (+audio) training including in GRPO. Let me fetch the ms-swift README to confirm native video AND audio training + custom reward plugin for video, and confirm the license. Let me also confirm the timestamp/temporal-grounding angle and verify licenses precisely.

18. user (2026-06-09T11:26:35.693Z)

Web search results for query: "ms-swift Apache license LLaMA-Factory Apache axolotl unsloth license trl Apache"

Links: [{"title": "Fine-tuning Tools Comparison | Guides | Clore.ai", "url": "https://docs.clore.ai/guides/comparisons/finetuning-comparison"}, {"title": "MS-SWIFT - Qwen", "url": "https://qwen.readthedocs.io/en/latest/training/ms_swift.html"}, {"title": "EVAL #003: Fine-Tuning in 2026 - Axolotl vs Unsloth vs TRL vs LLaMA-Factory - DEV Community", "url": "https://dev.to/ultraduneai/eval-003-fine-tuning-in-2026-axolotl-vs-unsloth-vs-trl-vs-llama-factory-2ohg"}, {"title": "GitHub - modelscope/ms-swift: Use PEFT or Full-parameter to CPT/SFT/DPO/GRPO 600+ LLMs (Qwen3.6, DeepSeek-V4, GLM-5.1, InternLM3, Llama4, ...) and 300+ MLLMs (Qwen3-VL, Qwen3-Omni, InternVL3.5, Ovis2.5, GLM4.5v, Gemma4, Llava, Phi4, ...) (AAAI 2025).", "url": "https://github.com/modelscope/ms-swift"}, {"title": "ms-swift · PyPI", "url": "https://pypi.org/project/ms-swift/"}, {"title": "GitHub - hiyouga/LlamaFactory: Unified Efficient Fine-Tuning of 100+ LLMs & VLMs (ACL 2024)", "url": "https://github.com/hiyouga/LlamaFactory"}, {"title": "Choosing the right LLM tuning library: Unsloth, Axolotl ...", "url": "https://www.linkedin.com/posts/aaditsh_confused-about-which-llm-tuning-library-to-activity-7369866434577002496-ZuYG"}, {"title": "Top 10 Open-Source Libraries to Fine-Tune LLMs Locally", "url": "https://www.analyticsvidhya.com/blog/2026/05/open-source-libraries-to-fine-tune-llm-locally/"}, {"title": "Unsloth vs Axolotl vs LLaMA-Factory - by Paolo Perrone", "url": "https://theaiengineer.substack.com/p/unsloth-vs-axolotl-vs-llama-factory"}]

Based on the search results, I found information about several fine-tuning frameworks for large language models, but the search results don't contain specific information about the Apache licenses for all the tools you mentioned. Here's what I found:

Frameworks Mentioned:

MS-SWIFT is the large model and multimodal large model training and deployment framework provided by the ModelScope community, and it appears to be available on GitHub as ms-swift.

LLaMA-Factory benefits from PEFT, TRL, QLoRA and FastChat, and it's a popular fine-tuning framework for LLMs.

The search results also discuss **Axolotl**, **Unsloth**, and **TRL** as other popular fine-tuning frameworks, but the specific Apache license details for each tool were not included in the search results I received.

To get accurate information about which of these tools are specifically licensed under the Apache License (and which versions), you would need to check each project's GitHub repository directly or look at their LICENSE files, as this licensing information wasn't included in the search results returned.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

19. user (2026-06-09T11:26:36.002Z)

ms-swift Software License and Capabilities

License

The software is licensed under **Apache License 2.0** (Apache-2.0). The repository states: "This framework is licensed under the Apache License (Version 2.0)."

Video and Audio Training Support

The documentation indicates support for "mixed modality data training with text, images, video and audio," suggesting capability for video and audio inputs. However, the page does not explicitly detail native video or audio-specific training implementations for models like Qwen3-Omni or Gemma.

GRPO with Custom Reward Functions

The framework supports "extensible reward functions" and "multi-turn inference Schedulers, and environments through plugins" for GRPO. However, specific documentation for multimodal video input support with custom external reward functions is not detailed on this page.

Video Frame Controls

The content provided does not mention per-frame timestamps, video frame sampling controls, fps settings, or num_frames parameters.

Supported Multimodal Models

Mentioned VLM models include:

- Qwen3-VL
- Qwen3-Omni
- InternVL3.5
- Gemma4
- GLM4.5v
- Llava
- Phi4
- Ovis2.5
- MiniCPM-V-4

Qwen2.5-VL is not explicitly listed in the provided content.

20. assistant (2026-06-09T11:26:39.474Z)

Let me confirm licenses precisely for each framework and check the timestamp/temporal grounding angle (whether these are framework features or just dataset formatting). The license question is critical given the hard commercial guardrail.

21. user (2026-06-09T11:26:48.102Z)

LLaMA Factory License & Capabilities

****License:**** Apache-2.0 License with no stated commercial-use restrictions. The README notes: "follow the model licenses to use the corresponding model weights" for individual models.

****VIDEO/AUDIO Training:**** Yes, LLaMA Factory supports multimodal fine-tuning. The changelog mentions "Qwen2-Audio" and "Qwen2.5-Omni" support for "audio understanding tasks." Video is supported via "LLaVA-NeXT-Video" models listed in supported models.

****GRPO/RL with Custom Rewards:**** The framework does not natively support custom reward functions for GRPO. The changelog states: "Announcing EasyR1, an efficient, scalable and multi-modality RL training framework for efficient GRPO training." This indicates GRPO functionality is delegated to the separate EasyR1 repository rather than implemented directly in LLaMA Factory.

22. user (2026-06-09T11:26:49.012Z)

Axolotl Multimodal Support Overview

VIDEO & AUDIO Training

****Video:**** The documentation explicitly states: "This is not well tested at the moment. We welcome contributors!" Video support exists but lacks maturity.

****Audio:**** Audio training is supported for specific models. The Voxtral model handles audio specifically, requiring installation of `librosa==0.11.0` and `mistral\_common[audio]==1.8.3`.

Models Supporting Video & Audio

- ****Audio:**** Voxtral Mini (requires special audio libraries)
- ****Video:**** SmolVLM2 is listed as "Video-Instruct" capable
- ****Gemma-4:**** Supports audio encoding alongside vision

Most other models focus on vision/images only.

Status

The overall feature is labeled ****(BETA)**** at the framework level. Video specifically has lower maturity with a community contribution call.

GRPO/RL with Custom Rewards

The documentation mentions "GRPO Training" and "Reward Modelling" as separate how-to guides, but the provided content doesn't detail custom reward function implementation specifics for VLMs.

Software License

The license information isn't included in this documentation page. You'd need to check the repository's LICENSE file or main README.

23. assistant (2026-06-09T11:26:53.071Z)

Important nuance: LLaMA-Factory's GRPO is delegated to EasyR1 (separate repo). Axolotl video is "not well tested." Let me confirm the licenses for axolotl, unsloth, TRL, and EasyR1, and verify the temporal-grounding/timestamp output angle is a model-recipe concern not a framework feature.

24. user (2026-06-09T11:26:58.808Z)

Web search results for query: "axolotl-ai-cloud license Apache unsloth license Apache OR AGPL EasyR1 license"

Links: [{"title": "Axolotl", "url": "https://docs.axolotl.ai/"}], [{"title": "Axolotl AI", "url": "https://docs.axolotl.ai/index.html"}], [{"title": "Unsloth: Strategic Open Source Alternative to OpenAI Fine-tuning", "url": "https://www.opentechhub.io/unsloth/"}], [{"title": "Open Source Licenses: Which One Should You Pick? MIT, GPL, Apache, AGPL and More (2026 Guide) - DEV Community", "url": "https://dev.to/juanisidoro/open-source-licenses-which-one-should-you-pick-mit-gpl-apache-agpl-and-more-2026-guide-p90"}], [{"title": "Introducing Unsloth Studio | Unsloth Documentation", "url": "https://unsloth.ai/docs/new/studio"}], [{"title": "Open Source Licensing Simplified: MIT, Apache, GPL, AGPL & More - The Tech Trends", "url": "https://thetechtrends.tech/open-source-licensing-models-guide/"}], [{"title": "GitHub - unslothai/unsloth: Unsloth Studio is a web UI for training and running open models like Gemma 4, Qwen3.6, DeepSeek, gpt-oss locally. · GitHub", "url": "https://github.com/unslothai/unsloth?locale=en-US"}], [{"title": "unsloth · PyPI", "url": "https://pypi.org/project/unsloth/"}], [{"title": "FAQ on Software Licensing | Elastic", "url": "https://www.elastic.co/pricing/faq/licensing"}], [{"title": "Understanding Open Source Licenses: GPL, MIT, Apache Compared - credativ®", "url": "https://www.credativ.de/en/blog/credativ-inside/understanding-open-source-licenses-gpl-mit-apache-compared/"}]

Based on the search results, here's the licensing information for the projects you asked about:

Axolotl-ai-cloud

Axolotl is licensed under the Apache 2.0 License.

Unsloth

Unsloth uses a dual-licensing model of Apache 2.0 and AGPL-3.0, where the core Unsloth package remains licensed under Apache 2.0, while certain optional components, such as the Unsloth Studio UI are licensed AGPL-3.0.

EasyR1

The search results did not return specific information about the licensing for a project called "EasyR1". You may want to search for this project directly or provide additional context if you're looking for a specific EasyR1 repository.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

25. user (2026-06-09T11:27:04.176Z)

Web search results for query: "ms-swift video temporal grounding timestamp fine-tuning Qwen2.5-VL fps num_frames training dataset format"

Links: [{"title": "Video Understanding and Grounding with Qwen 2.5 - PyImageSearch", "url": "https://pyimagesearch.com/2025/06/16/video-understanding-and-grounding-with-qwen-2-5/"}], [{"title": "TEMPURA: Temporal Event Masked Prediction and Understanding for Reasoning in Action", "url": "https://arxiv.org/pdf/2505.01583"}], [{"title": "Temporal Grounding · Issue #752 · QwenLM/Qwen2.5-VL", "url": "https://github.com/QwenLM/Qwen2.5-VL/issues/752"}], [{"title": "Know-Show: Benchmarking Video-Language Models on Spatio-Temporal Grounded Reasoning", "url": "https://arxiv.org/pdf/2512.05513"}], [{"title": "Qwen3-VL Technical Report", "url": "https://arxiv.org/pdf/2511.21631"}], [{"title": "Support for finetuning of Qwen2-VL-Chat series models · Issue #1857 · modelscope/ms-swift", "url": "https://github.com/modelscope/ms-swift/issues/1857"}], [{"title": "MUSEG: Reinforcing Video Temporal Understanding via Timestamp-Aware Multi-Segment Grounding", "url": "https://arxiv.org/pdf/2505.20715"}], [{"title": "About Qwen2.5 VL finetuning · Issue #5280 · modelscope/ms-swift", "url": "https://github.com/modelscope/ms-swift/issues/5280"}], [{"title": "ms-swift-2.5.0.post1-241017/docs/source_en/Multi-Modal/qwen2-vl-best-practice.md at main · xuyongfu/ms-swift-2.5.0.post1-241017", "url": "https://github.com/xuyongfu/ms-swift-2.5.0.post1-241017/blob/main/docs/source_en/Multi-Modal/qwen2-vl-best-practice.md"}], [{"title": "TempSamp-R1: Effective Temporal Sampling with Reinforcement Fine-Tuning for Video LLMs", "url": "https://arxiv.org/pdf/2509.18056"}]

Based on the search results, here's what I found regarding video temporal grounding, timestamp

fine-tuning, and Qwen2.5-VL:

Temporal Grounding and Timestamp Handling

Unlike traditional methods that rely on textual timestamps or add computational layers to enable temporal grounding, Qwen2.5 employs an innovative approach by aligning Multimodal Rotary Position Embedding (MRoPE) IDs with timestamps, allowing the model to grasp the tempo of time directly through intervals between temporal dimension IDs.

However, the original temporal encoding scheme of Qwen2.5-VL tends to misalign for fine-grained temporal grounding, especially in longer videos. To address this, visual timestamps can be overlaid on the upper left corner of each sampled video frame to explicitly mark the temporal context.

Dynamic FPS and num_frames Training

Dynamic Frames Per Second (FPS) sampling during training ensures a well-distributed representation of frame rates within the dataset. For example, all videos are temporally downsampled to 2 frames per second (FPS) and resized to approximately 2.8 million pixels per frame.

Training Dataset Format

The fine-tuning data format used with ms-swift follows a JSON structure with "messages" containing role-based content, along with "images" and optional "objects" fields containing references and bounding box coordinates.

Alternative Approaches

In newer approaches, a textual token-based time encoding strategy is adopted, wherein each video temporal patch is prefixed with a timestamp expressed as a formatted text string?e.g., <3.0 seconds>, and during training, timestamps are generated in both seconds and HMS (hours:minutes:seconds) formats.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

26. assistant (2026-06-09T11:27:13.902Z)

I now have enough to confirm the key finding: temporal grounding/timestamp outputs are a model+dataset-format concern (Qwen2.5-VL MRoPE timestamps), and ms-swift exposes the fps/num_frames controls and video dataset format to train it. The RFT-with-custom-reward path for video (e.g. timestamp-aware multi-segment grounding via GRPO) is exactly what ms-swift's external reward plugin supports natively for video, whereas TRL/Unsloth GRPO-VLM is image-only and LLaMA-Factory delegates to EasyR1.

Let me do one final confirmation on ms-swift's GRPO for video specifically (custom reward + video rollouts) and the audio (Omni) training, plus confirm EasyR1 license, to lock down the comparison.

27. user (2026-06-09T11:27:19.907Z)

EasyR1 License and Features Analysis

****License:**** Apache-2.0 (as stated in the repository)

****Video VLM GRPO Training:**** The documentation makes no mention of video capabilities. The supported models are limited to text and image-based vision language models.

****Custom Reward Functions:**** There is no evidence in the provided content that EasyR1 supports custom reward functions for GRPO training.

****Supported VLM Models:**** The framework explicitly supports "Qwen2-VL/Qwen2.5-VL/Qwen3-VL vision language models" alongside language-only models like Llama and Qwen variants.

****Audio Support:**** The repository contains no references to audio model support. The framework is designed for text and vision-language tasks only.

The documentation emphasizes that EasyR1 is "a clean fork of the original veRL project to support vision language models" with focus on image-text and multi-image-text datasets, but provides no indication of video or audio modality support.

28. user (2026-06-09T11:27:24.750Z)

Web search results for query: "ms-swift GRPO video reward_funcs external_plugin Qwen2.5-VL rollout vLLM multimodal example"

Links: [{"title": "ms-swift - Qwen", "url": "https://qwen.readthedocs.io/en/v2.5/training/RL/ms_swift.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.4.1.post1 documentation", "url": "https://swift.readthedocs.io/en/v3.4/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 4.1.0.dev0 documentation", "url": "https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.11.3 documentation", "url": "https://swift.readthedocs.io/en/v3.11/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.7.3 documentation", "url": "https://swift.readthedocs.io/en/v3.7/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.5.3 documentation", "url": "https://swift.readthedocs.io/en/v3.5/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.9.3 documentation", "url": "https://swift.readthedocs.io/en/v3.9/BestPractices/GRPO-Multi-Modal-Training.html"}, {"title": "Complete Multimodal GRPO Experiment Workflow ? swift 3.6.4 documentation", "url": "https://swift.readthedocs.io/en/v3.6/BestPractices/GRPO-Multi-Modal-Training.html"}]

The search results contain several examples and documentation related to your query. Here's what I found:

MS-SWIFT supports GRPO training with Qwen2.5-VL models using external plugins and reward functions, with vLLM enabled for inference acceleration during rollout.

The documentation shows several multimodal GRPO training examples. For relatively simple tasks, `max_completion_length` is set to 1024 and `external_r1v_acc` and `format` are selected as reward functions.

Several example configurations are provided:

1. A multi-GPU setup (8 GPUs) using Qwen2.5-VL-7B-Instruct with `external_plugins`, `reward_funcs` set to `external_r1v_acc` and `format`, vLLM enabled with 8 inference workers, and the multimodal-open-r1-8k-verified dataset.
2. A 6-GPU configuration using Qwen2.5-VL-3B-Instruct with the same `external_plugins` and `reward_funcs` setup, trained on the `clevr_cogen_a_train` or `GEQA_R1V_Train_8K` datasets.

The examples consistently use:

- `--external_plugins examples/train/grpo/plugin/plugin.py` for custom reward implementations
- `--reward_funcs external_r1v_acc format` for the reward functions
- `--use_vllm true` to enable vLLM for efficient rollout generation
- Various datasets including multimodal reasoning benchmarks

These configurations provide a complete workflow for multimodal GRPO training with vLLM-based rollout generation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

29. assistant (2026-06-09T11:27:34.806Z)

I now have a complete, verified picture. Let me consolidate the findings. The decisive points:

1. **ms-swift** ? Apache-2.0, the only framework that natively trains video AND audio (Qwen3-Omni/Gemma4), with GRPO + custom external reward plugins that work on multimodal/video rollouts via vLLM, plus video frame-sampling controls (fps/num_frames) for timestamp grounding. This is the clear primary.
2. **LLaMA-Factory** ? Apache-2.0, native video+audio SFT, but GRPO is delegated to the separate **EasyR1** (Apache-2.0) which is **image-only** (no video, no audio, no custom-reward video path confirmed). So the RFT-with-custom-reward-on-video requirement is not met in one stack.
3. **Unsloth** ? Apache-2.0 core (Studio UI is AGPL, avoidable), single-GPU optimized, but its Vision-RL/GRPO is image-only; great for the 8GB box but doesn't fit the video-RFT requirement.
4. **Axolotl** ? Apache-2.0, but video is explicitly "not well tested"; weak fit for a video-first quality goal.
5. **TRL** ? Apache-2.0, the substrate library; GRPO-VLM is image-only in cookbooks; you'd build the video plumbing yourself.

Framework vs base model orthogonality is clear to flag. Let me also verify the temporal-grounding point is framework-agnostic before finalizing ? it is (it's Qwen MRoPE + dataset format, which ms-swift exposes). I have what I need.

30. user (2026-06-09T11:28:55.330Z)

Output does not match required schema: root: must have required property 'confidence'

31. assistant (2026-06-09T11:28:58.829Z)

The `confidence` field got mis-parsed (a tag-name escaping issue). Let me resubmit with all fields correctly formed.

32. user (2026-06-09T11:29:55.297Z)

Structured output provided successfully

33. assistant (2026-06-09T11:30:11.832Z)

Structured output delivered successfully. Summary of the framework recommendation for the owner:

Decision: adopt ms-swift (Apache-2.0) as the single SFT+RFT forge now.

The orthogonality clarification the owner asked for: framework (the trainer/forge) and base model (Qwen/Gemma) are independent picks ? you choose one framework and feed it any Apache/MIT base, and swapping the base later is a config change, not a re-platform.

Why ms-swift wins decisively: it is the *only* one of the five that meets all four hard requirements in a single stack ? native video, native audio (Qwen3-Omni/Gemma-4), timestamp/temporal-grounding controls (fps/num_frames + the JSON timestamped-messages format), and GRPO/RFT with a *custom external-reward plugin run over video rollouts* (vLLM). Its reward-plugin seam is exactly where your existing `metrics.match\_intervals` temporal-IoU scorer drops in as the RFT reward.

Why each competitor falls short:

- **LLaMA-Factory** (Apache-2.0): native video+audio SFT, but offloads GRPO to **EasyR1**, which is image-only ? so video-RFT-with-custom-reward isn't achievable in that stack.
- **Unsloth** (Apache-2.0 core; AGPL only in the optional Studio UI): its Vision-RL is image-only ? but it is the **best fit for the 8GB RTX 2070 Super**, so keep it as the secondary tool for the *deferred* on-device/compression goal.
- **TRL** (Apache-2.0): the cleanest custom-reward API and the substrate everything else builds on, but GRPO-VLM is image-only; video plumbing is DIY. Keep it in view as the portability hedge.
- **Axolotl** (Apache-2.0): video is self-described as "not well tested" ? wrong bet for a video-first quality goal.

Two framing points folded into the output: (1) the commercial-license guardrail is **not**

decided at the framework layer (all five are permissive) ? it's decided by base-weight provenance (your open WASB C3 blocker) and training data (no paid-LLM labels); (2) conversational-QA bookkeeping is an application/retrieval layer no trainer holds, so the framework choice doesn't force the embark-now-vs-punt decision ? pick the forge now, build the per-video state store as the brief requires, and add QA as an SFT task to ms-swift later.

This hardens the existing lean in `docs/PLATFORM\_ARCHITECTURE.md` (§"Training backend" and §"Training harness"), which already favored ms-swift but predated the RFT-custom-reward and audio requirements that now make the gap decisive.